

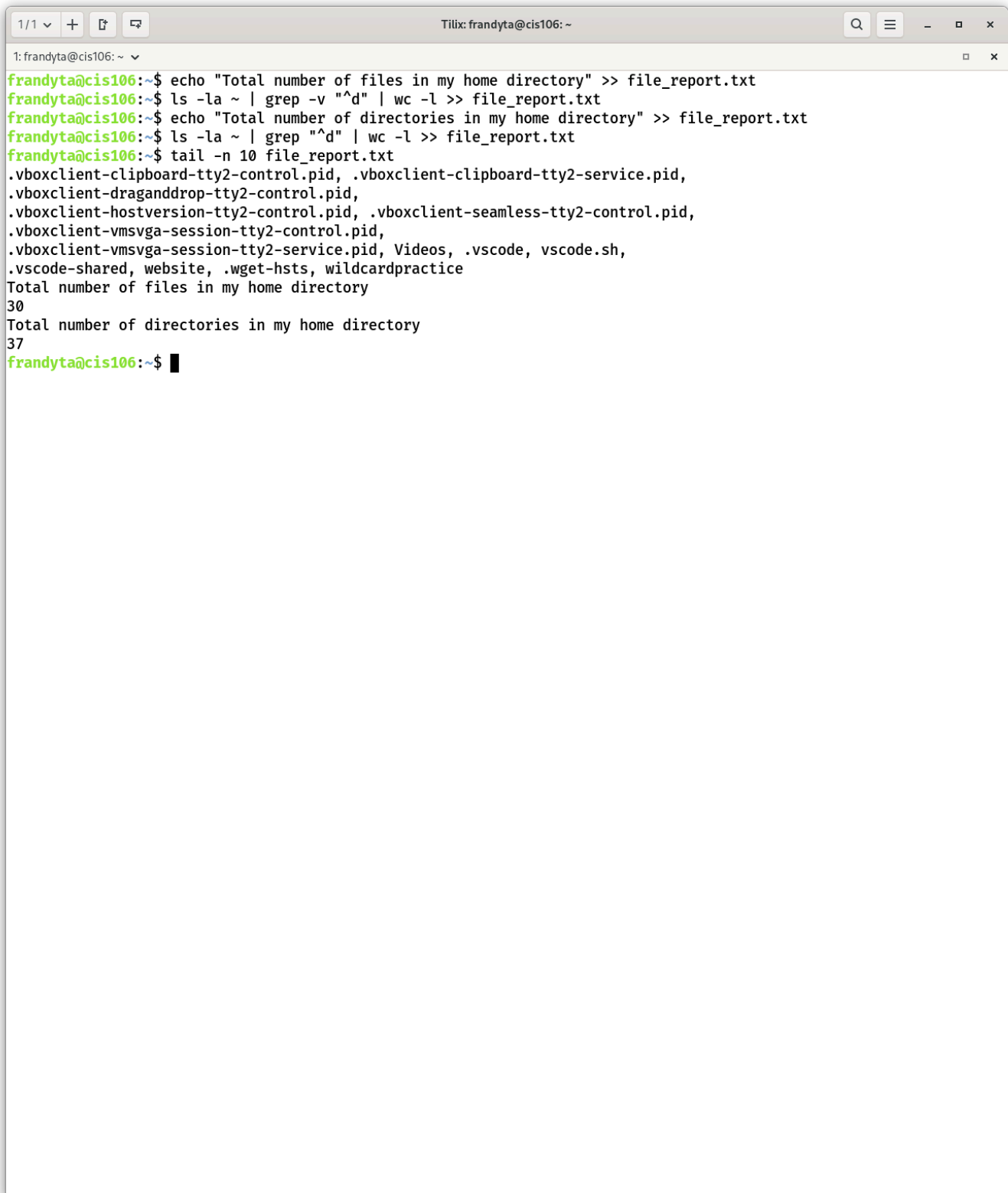
Lab 9

Question 1

A terminal window titled 'Tilix: frandyta@cis106: ~' showing a series of commands and their output. The user 'frandyta' is at the 'cis106' machine. They first echo a message into a file named 'file_report.txt'. Then, they use 'ls -la ~' to list all files in their home directory, redirecting the output to 'file_report.txt'. Finally, they use 'cat file_report.txt' to display the contents of the file. The output lists various files and directories, including hidden files like '.bashrc' and '.ssh', and regular files like 'Hello.txt' and 'man-page-activity.pdf.1'. The terminal window has a standard Ubuntu-style title bar and window controls.

```
1/1 | + | |  
Tilix: frandyta@cis106: ~  
1: frandyta@cis106: ~  
frandyta@cis106:~$ echo "All files in my home directory" > file_report.txt  
frandyta@cis106:~$ ls -la ~ >> file_report.txt  
frandyta@cis106:~$ cat file_report.txt  
All files in my home directory  
., .., access.log, .bash_aliases, .bash_history, .bash_logout, .bashrc,  
blue.txt, .cache, cis106, .config, Desktop, Documents, .dotnet, Downloads,  
essentials.sh, .face, .face.icon, file_report.txt, . Fonts, .gitconfig, .gnupg,  
Hello.txt, hmw, inClassActivity, lab5files, lab6-challenge1, lab6-challenge2,  
.local, man-page-activity.pdf, man-page-activity.pdf.1, marstenHouse,  
marstenhouse.sh, movies, .mozilla, music, Music, My Vacation 2026, Pictures,  
.pki, practicels, practicels.tar.xz, practicels.tar.xz.1, .profile, Public,  
Script, snap, sorted_access_log_by_endpoint.log, sorted_access_log_by_username.log,  
.ssh, .sudo_as_admin_successful, Templates,  
.vboxclient-clipboard-tty2-control.pid, .vboxclient-clipboard-tty2-service.pid,  
.vboxclient-draganddrop-tty2-control.pid,  
.vboxclient-hostversion-tty2-control.pid, .vboxclient-seamless-tty2-control.pid,  
.vboxclient-vmvga-session-tty2-control.pid,  
.vboxclient-vmvga-session-tty2-service.pid, Videos, .vscode, vscode.sh,  
.vscode-shared, website, .wget-hsts, wildcardpractice  
frandyta@cis106:~$
```

Question 2

A terminal window titled 'Tilix: frandyta@cis106: ~' showing a series of shell commands and their outputs. The commands are used to count files and directories in the home directory and list the last 10 lines of a report file. The output shows 30 files and 37 directories.

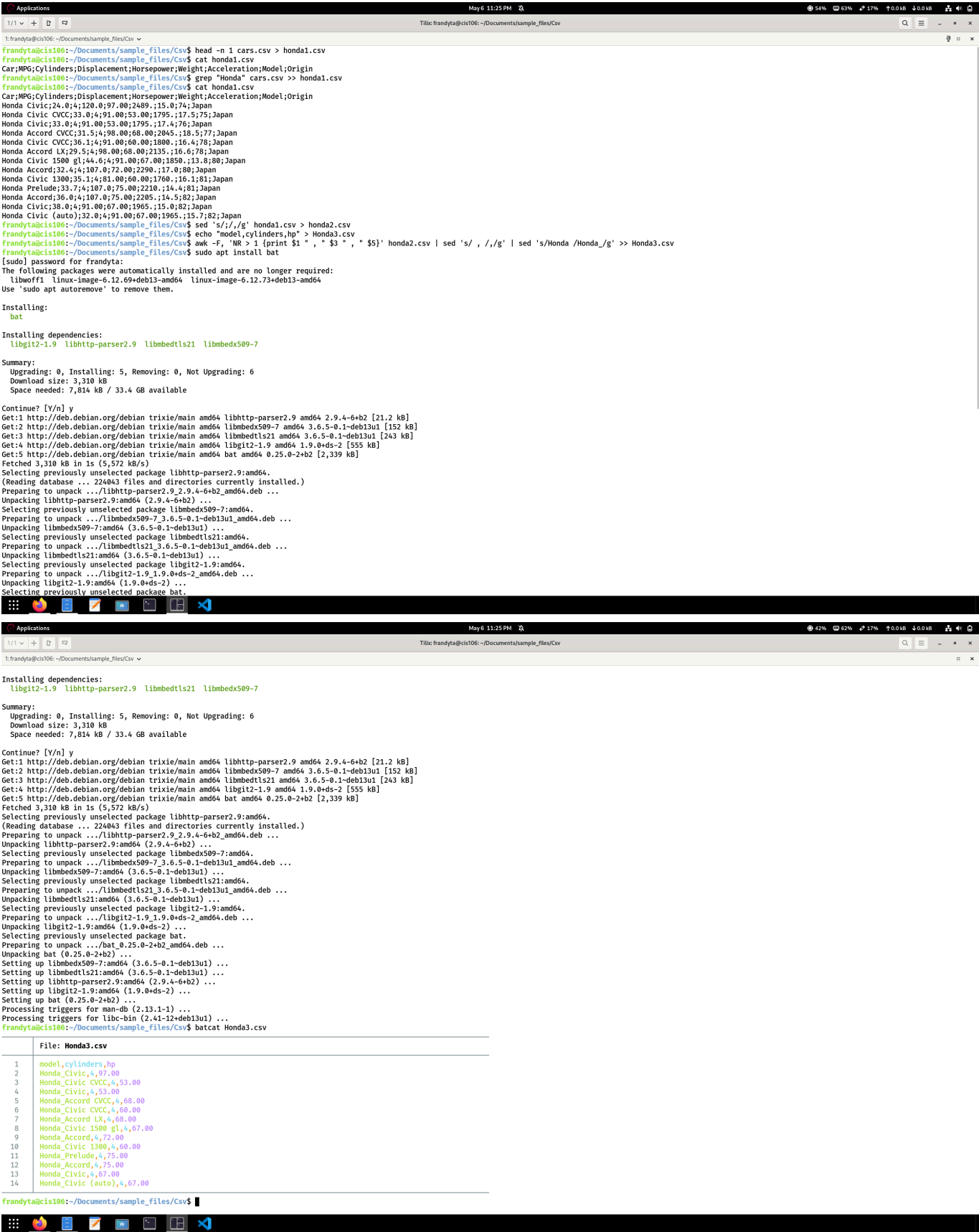
```
1/1 v + [ ] [ ]
Tilix: frandyta@cis106: ~
1: frandyta@cis106: ~
frandyta@cis106:~$ echo "Total number of files in my home directory" >> file_report.txt
frandyta@cis106:~$ ls -la ~ | grep -v "^d" | wc -l >> file_report.txt
frandyta@cis106:~$ echo "Total number of directories in my home directory" >> file_report.txt
frandyta@cis106:~$ ls -la ~ | grep "^d" | wc -l >> file_report.txt
frandyta@cis106:~$ tail -n 10 file_report.txt
.vboxclient-clipboard-tty2-control.pid, .vboxclient-clipboard-tty2-service.pid,
.vboxclient-draganddrop-tty2-control.pid,
.vboxclient-hostversion-tty2-control.pid, .vboxclient-seamless-tty2-control.pid,
.vboxclient-vmvga-session-tty2-control.pid,
.vboxclient-vmvga-session-tty2-service.pid, Videos, .vscode, vscode.sh,
.vscode-shared, website, .wget-hsts, wildcardpractice
Total number of files in my home directory
30
Total number of directories in my home directory
37
frandyta@cis106:~$
```

Question 3

```
Applications May 6 11:14 PM 40% 57% 18% 0.1kB 0.1kB
Tilix: frandyta@cis106: ~
frandyta@cis106:~$ awk -F: '{print $1}' /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
dhcpcd
tss
systemd-timesync
messagebus
dnsmasq
avahi
speech-dispatcher
usbmux
cups-pk-helper
fwupd-refresh
geoclue
gnome-remote-desktop
saned
polkitd
rtkit
colord
Debian-gdm
frandyta
_flatpak
vboxadd
frandyta@cis106:~$ awk -F: '{print $1, $7}' /etc/passwd
root /bin/bash
daemon /usr/sbin/nologin
bin /usr/sbin/nologin
sys /usr/sbin/nologin
sync /bin/sync
games /usr/sbin/nologin
man /usr/sbin/nologin
lp /usr/sbin/nologin
mail /usr/sbin/nologin
news /usr/sbin/nologin
uucp /usr/sbin/nologin
proxy /usr/sbin/nologin
www-data /usr/sbin/nologin
backup /usr/sbin/nologin
list /usr/sbin/nologin
irc /usr/sbin/nologin
_apt /usr/sbin/nologin
nobody /usr/sbin/nologin
systemd-network /usr/sbin/nologin
Debian-gdm
frandyta
_flatpak
vboxadd
frandyta@cis106:~$ awk -F: '{print $1, $7}' /etc/passwd
root /bin/bash
daemon /usr/sbin/nologin
bin /usr/sbin/nologin
sys /usr/sbin/nologin
sync /bin/sync
games /usr/sbin/nologin
man /usr/sbin/nologin
lp /usr/sbin/nologin
mail /usr/sbin/nologin
news /usr/sbin/nologin
uucp /usr/sbin/nologin
proxy /usr/sbin/nologin
www-data /usr/sbin/nologin
backup /usr/sbin/nologin
list /usr/sbin/nologin
irc /usr/sbin/nologin
_apt /usr/sbin/nologin
nobody /usr/sbin/nologin
systemd-network /usr/sbin/nologin
dhcpcd /bin/false
tss /bin/false
systemd-timesync /usr/sbin/nologin
messagebus /usr/sbin/nologin
dnsmasq /usr/sbin/nologin
avahi /usr/sbin/nologin
speech-dispatcher /bin/false
usbmux /usr/sbin/nologin
cups-pk-helper /usr/sbin/nologin
fwupd-refresh /usr/sbin/nologin
geoclue /usr/sbin/nologin
gnome-remote-desktop /usr/sbin/nologin
saned /usr/sbin/nologin
polkitd /usr/sbin/nologin
rtkit /usr/sbin/nologin
colord /usr/sbin/nologin
Debian-gdm /bin/false
frandyta /bin/bash
_flatpak /usr/sbin/nologin
vboxadd /bin/false
frandyta@cis106:~$ awk -F: '{print $1, $7, $6}' /etc/passwd
root /bin/bash /root
daemon /usr/sbin/nologin /usr/sbin
bin /usr/sbin/nologin /bin
sys /usr/sbin/nologin /dev
sync /bin/sync /bin
games /usr/sbin/nologin /usr/games
man /usr/sbin/nologin /var/cache/man
lp /usr/sbin/nologin /var/spool/lpd
mail /usr/sbin/nologin /var/mail
news /usr/sbin/nologin /var/spool/news
uucp /usr/sbin/nologin /var/spool/uucp
proxy /usr/sbin/nologin /bin
www-data /usr/sbin/nologin /var/www
backup /usr/sbin/nologin /var/backups
list /usr/sbin/nologin /var/list
```



Question 4



Challenge Question

